Execution-Constrained Portfolio Rebalancing: Policy Domains, Search and Numerical Reliability

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Abstract

Executing a target portfolio requires choosing actual trades under funding, exposure, minimum-size and ticket constraints. We specify these choices as labelled executable intervals, separating compulsory actions, common feasibility preparation and tracking-risk minimisation. A correlated two-asset example shows that restricting movements to target corridors can raise annual tracking error from 43.59 to 100 basis points. We compare joint mixed-integer optimisation (Exact Execution), bounded support search (Ranked Execution) and a documented Kernel Search adaptation on the same prepared domain. A retained 313-case cohort favours Exact's median computation, but a frozen prospective study reveals different limits. On 18 larger-candidate synthetic cases, Exact and Kernel return audited portfolios in every call; Ranked returns two within its declared process ceiling. Kernel improves Exact's 60-second incumbent in 11 cases, ties four and loses three. All three 600-second Exact controls remain unresolved. Two larger externally sourced risk cases instead favour Exact. Equivalent numerical controls expose original-unit bound inconsistencies despite native optimal labels; balanced representations avoid those contradictions in the tested cases. The evidence distinguishes economic restrictions, search quality and numerical acceptance. It supports conditional execution comparisons, not universal method superiority or realised investment-performance claims.

Keywords: portfolio rebalancing, trade cardinality, tracking error, mixed-integer quadratic optimisation, kernel search, numerical reliability

1. Introduction

Selecting movements from an existing portfolio differs from selecting the assets in a new portfolio. A limited order list may need to fund purchases, exit ineligible holdings and respect overlapping exposures. The value of a trade depends on the other trades that can accompany it. Minimum delivered orders and minimum final holdings further restrict those combinations.

This paper asks how the declared execution domain, support search and numerical representation affect executable delivery. The objective is tracking variance against a supplied raw target. Alpha, turnover penalties, transaction costs and upstream target estimation are outside that execution objective. We compare methods only after resolving the same mandatory actions and feasible base.

Our contribution has three parts. First, an explicit state formulation separates trade-count semantics, preparation and economic restrictions; an independent counterexample establishes that target corridors need not preserve the full-domain optimum. Second, controlled representations distinguish the mathematical objective from native stopping diagnostics and an independently audited delivery. Third, frozen supplied-input experiments identify operating limits of three declared implementations, including failures and outcomes against a preferred method. The disjunctive and perspective constructions are established tools, and the primary/fallback wrapper is not claimed as a new optimisation principle.

The evidence is deliberately not a single ranking. Exact is computationally attractive in the retained small-candidate cohort. Kernel frequently finds better incumbents in the new synthetic stress cells, but loses in other cells and in the larger external-risk cases. Ranked often returns no result within its process ceiling. These observations motivate an explicit separation of policy-domain costs, search limits and numerical reliability.

2. Related work

Index tracking and sparse portfolio selection provide the closest optimisation context. Beasley et al. (2003) study an evolutionary tracking heuristic; Canakgoz and Beasley (2009) develop mixed-integer tracking and enhanced-indexation models. Mutunge and Haugland (2018) examine cardinality-constrained tracking portfolios, and Strub and Baumann (2018) study their construction and rebalancing. Our cardinality counts movements from an

existing book rather than terminal holdings. Hard execution rules also differ from a trading-cost objective, although fixed costs and minimum orders belong to the established nonconvex portfolio-modelling family described by Lobo et al. (2007).

Finite unions of executable intervals admit standard disjunctive extended formulations (Vielma, 2015). Perspective methods strengthen quadratic relaxations with indicator variables (Frangioni and Gentile, 2006; Günlük and Linderoth, 2010). We use those properties without claiming that the coupled portfolio relaxation is ideal. The ridge-regularised sparse model of Bertsimas and Cory-Wright (2022) provides another scalable approach, but adding its regulariser would change the objective here and therefore would not constitute a matched comparator.

Kernel Search explores restricted mixed-integer subproblems (Guastaroba and Speranza, 2012, 2014). Our comparator adapts that framework to the same quadratic executable-state model, and explicitly describes departures from the published algorithms. Ranked Execution instead constructs and replaces supports, then jointly sizes them. Neither restricted-support search nor support exchange is new. The empirical question is how these particular implementations behave under identical economic inputs and independent starts.

3. Execution domain and policy restrictions

3.1. Inputs and constraints

Let $c, m, w \in \mathbb{R}^N$ be original current, raw target and delivered weights, measured as fractions of net asset value (NAV). Settlement cash has index k. With supplied annual covariance $\Sigma \succeq 0$, minimise

$$f(w) = (w - m)^\top \Sigma (w - m), \qquad \mathrm{TE}(w) = \sqrt{f(w)}. \tag{1}$$

Annual tracking-error basis points are $10^4\,\mathrm{TE}(w)$. The long-only hard domain is

$$\mathbf{1}^\top w = 1, \qquad L \le w \le U, \qquad a \le Aw \le b. \tag{2}$$

The exposure matrix A can contain overlapping and fractional memberships. Cash has no ticket, but retains bounds and covariance contributions.

An ordinary noncash movement is zero or at least δ_i in absolute size; a final holding is zero or at least θ_i. These are different rules:

$$w_i = c_i \text{ or } |w_i - c_i| \ge \delta_i, \qquad w_i = 0 \text{ or } w_i \ge \theta_i. \tag{3}$$

Tickets count actual delivered movements. An explicit compulsory minimum-size exemption does not remove its ticket or waive its holding floor.

Table 1: Principal notation and separate numerical conventions.

Symbol	Meaning
c, m, q, w	Original current, raw target, common prepared base, delivered weights.
N, k, Σ	Number of coordinates including cash, cash index, annual covariance.
M, B, K	Reserved mandatory names, optional allowance, total ticket capacity.
$\ell_{ij}, u_{ij}, t_{ij}$	Executable interval endpoints and ticket label.
δ_i, θ_i	Minimum ordinary trade and minimum positive holding.
$10^{-8}, 10^{-7}$	Trade-zero threshold and portfolio audit tolerance, in NAV fractions.
10^{-12}	Objective comparison scale, in annual variance.
10^{-4} bp	Descriptive TE equality band; not a feasibility tolerance.

Before solving, reconcile every declared group with the compiled rows. A missing membership column is an input error, not a zero vector. An explicitly acknowledged empty group is redundant only when its bounds contain zero; otherwise it is contradictory. Reconciliation records source identifiers, coefficients and bounds. It addresses a problem a return audit alone cannot detect: an omitted constraint would also be absent from that audit. No coefficient or tolerance is changed.

3.2. Mandatory actions and common preparation

The consumer resolves lifecycle changes, instructions and ordinary hard-bound repairs into a proposed base. Every genuine noncash base–current movement has an explicit mandatory category. A compulsory action can pin a weight or retain an admissible sizing interval; its original movement is counted in either case. Numerical residuals within the unchanged zero threshold do not create a new compulsory action.

Individual instructions may violate joint funding or exposures. Common preparation first retains an already audited base unchanged. Otherwise it searches the corridor domain and then eligible full bounds. In the

full-domain stage it minimises corridor excursion before gross noncash intervention and the number of added repair instruments. A strictly convex squared-intervention objective breaks continuous ties within the selected discrete pattern. Target tracking risk does not choose repairs. Sequential numerical guards qualify this priority order; it is not exact symbolic lexicographic arithmetic.

Additional repairs are pinned, categorised and counted before risk search. All comparison arms receive the same resulting q, bounds, exemptions and mandatory declarations. An optional allowance B gives $K = M + B$ under that explicit convention. An independently specified fixed total cap must instead remain fixed; preparation cannot silently enlarge it. Preparation failures stay in the denominator and are not automatically proofs of infeasibility.

3.3. Corridor versus full hard domain

For an ordinary optional name, a target corridor is

$$[\min(q_i, m_i), \max(q_i, m_i)] \cap [L_i, U_i].$$

Resolved instruction and compulsory intervals take precedence. The *Full* domain uses eligible hard bounds without the ordinary corridor restriction. Our algorithm comparisons use the same *Corridor* domain. Full comparisons are explicitly different policy problems.

A corridor can exclude the best execution combination. Let cash be the third coordinate, take one noncash ticket, bounds $[0, 1]$, minimum trade 0.005 and holding floor 0.0025, and set

$$c = (.2, .2, .6), \quad m = (.3, .3, .4), \quad \Sigma = .01 \begin{pmatrix} 1 & .9 & 0 \\ .9 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

If only the first risky asset trades, $x = w_1 - .3$ gives $f(w) = .01(x^2 - .18x + .01)$. The corridor minimum is $x = 0$, yielding $w = (.3, .2, .5)$ and TE 100 bp. Full permits $x = .09$, yielding $w = (.39, .2, .41)$ and TE 43.588989 bp. Trading the second asset is symmetric; no trade is worse. These scalar cases exhaust the ticket choices independently. The overshoot offsets an untraded correlated underweight. Thus the corridor is an economic policy restriction, not a generally lossless reduction.

4. Formulation and solution methods

4.1. Labelled executable states

For each noncash name i, compile intervals $[\ell_{ij}, u_{ij}]$, $j \in J_i$, by intersecting permitted movements with the hard bounds and the trade/holding rules. The no-trade state is a point with label $t_{ij} = 0$; active states have $t_{ij} = 1$. A full exit can be a separate state. Mandatory declarations remove inadmissible no-trade states. Let I_O denote optional names.

Introduce binary selectors z_{ij} and contributions v_{ij}. Exact Execution is the joint mixed-integer quadratic model for (1) and (2), with

$$\sum_{j \in J_i} z_{ij} = 1, \qquad \ell_{ij} z_{ij} \le v_{ij} \le u_{ij} z_{ij}, \qquad w_i = \sum_{j \in J_i} v_{ij}, \tag{4}$$

$$\sum_{i \neq k} \sum_{j \in J_i} t_{ij} z_{ij} \le K, \qquad \sum_{i \in I_O} \sum_{j \in J_i} t_{ij} z_{ij} \le B. \tag{5}$$

An unspecified capacity row is omitted. Exactly one selected interval supplies each weight; all inactive contributions are zero. Conversely every admissible labelled portfolio supplies such selectors and contributions. This proves projection equivalence with the declared domain, using a standard extended formulation (Vielma, 2015). It does not prove ideality of the coupled continuous relaxation. The supplement gives the complete argument and separates this nominal domain from finite-precision acceptance.

4.2. Exact, Ranked and Kernel Search

Exact Execution solves the joint state and sizing model. "Exact" identifies that formulation, not a promise that every time-limited call proves optimality. Its evaluated default is non-perspective. The frozen native controls retain one thread, relative-gap target 10^{-4} and an absolute target equivalent to 10^{-12} original variance. Audited callback incumbents are retained, and a bounded fixed-state refinement can repair a numerical boundary candidate. Refinement stays inside the same constraints and audit envelope.

Ranked Execution is a bounded support search. It computes cash-funded scalar and pair priorities, forms candidate supports and jointly sizes their selected states under all hard rows. Search includes mandatory-only sizing, greedy construction, repair supports, single replacements, conditional coordinated moves, perspective-guided restarts and screened three-position combinations. A cache avoids repeating support sizing. Identifier and state

order resolve ties. The released rescue budget is 24, mandatory-assignment budget 32 and improvement-pass budget three. The supplement gives the ordered pseudocode, including branch budgets and stopping; no new stopping rule is introduced here.

Perspective guidance splits the funded covariance as $Q = Q_0 + \mathrm{diag}(d)$ with $Q_0 \succeq 0$ and $d \geq 0$. For optional occupancy y_i, replace $d_i x_i^2$ by $d_i x_i^2 / y_i$ with its closed convex extension at zero. It equals the original quadratic at integral occupancy and strengthens the continuous relaxation (Frangioni and Gentile, 2006; Günlük and Linderoth, 2010). An approximate conic primal value alone is not a lower bound. Ranked's separately corrected relaxation-bound routine and native mixed-integer bound diagnostics are distinct mechanisms.

Kernel Search uses the same state formulation, independent prepared start, objective and audit. Following the framework of Guastaroba and Speranza (2012, 2014), our adaptation ranks names by fractional trade occupancy in the perspective relaxation, selects an initial kernel of at most $\max(24, 2B)$ names and partitions the remainder into equally sized buckets. Restricted joint solves retain all mandatory actions. A bucket solve requires a bucket trade and uses an incumbent objective cutoff; selected bucket names join the kernel. The remaining allowance is divided among remaining stages. One pass is made, without ageing or removal. Quadratic guidance, occupancy ordering and instrument-level state groups are explicit adaptations, not a literal replication of the original tracking model. Restricted subproblem bounds are never presented as global bounds.

The methods therefore use comparable economic problems but different search architectures. The study does not compare the strongest attainable versions of these algorithm families. For T distinct support trials and at most H mandatory assignments, Ranked attempts at most TH fixed-state sizing problems, in addition to guidance. Shortlist generation and dense covariance operations also cost time. There is no proved $O(NB)$ or approximation-ratio guarantee. Factor-plus-diagonal inputs do not make this implementation linear-memory: funded model construction and QIS reporting materialise dense matrices.

4.3. Audited execution policy

A separate operational policy attempts Exact first and calls Ranked only when the primary cannot provide an acceptable native candidate. It checks original inputs, callback/native failures and available bound consistency. The

prepared base alone is not evidence of successful native search. Fallback must pass the same audit, or the policy fails. This rule does not run both methods and select the better objective. Its preparation still depends on the common solver infrastructure, so it is not independent protection against a backend outage. The older operational MIQP has different policy/objective semantics and is not a matched arm in this paper.

5. Numerical representation and acceptance

Eliminate fixed coordinates and cash using a funded reference r and columns $e_i - e_k$ in D. Then $w = r + Dx$ and

$$f(r + Dx) = x^\top Q x + 2g^\top x + f(r), \quad Q = D^\top \Sigma D, \quad g = D^\top \Sigma (r - m). \tag{6}$$

All instrument and group rows, including cash bounds, undergo the same substitution. Funding elimination preserves the full covariance objective.

5.1. Objective origin and native bounds

A positive affine transformation $\alpha f + \beta$ preserves the minimiser, but changes the relative-gap denominator. For incumbent value F_U and lower bound F_L and denominator floor $\eta > 0$, the transformed gap has the form

$$\frac{\alpha |F_U - F_L|}{\max(\eta, |\alpha F_U + \beta|)}.$$

For $F_U = 10^{-10}$, $F_L = 9 \times 10^{-11}$, removing a constant 10^{-3} changes the relative gap from 0.1 to approximately 10^{-8}. The absolute gap 10^{-11} still exceeds 10^{-12}. This analytical pair shows why restoring only the minimiser-equivalent expression is insufficient for relative stopping. It is not fabricated native solver output. The implementation preserves the constant in (6) and restores variance units in diagnostics.

Absolute or relative native stopping may terminate a call (MOSEK ApS, 2026). A native optimal label, a feasible delivery and an independently established optimum remain different claims. Moreover, a better native vector can fail portfolio acceptance. Its native gap cannot be attributed to a different, worse audited portfolio. We retain both objectives and gaps, and check native bounds against all available audited witnesses in the same domain.

5.2. Equivalent cones and fixed rows

In the scaled objective, basis-point coordinates x_{bp} give the identity

$$(x_{\mathrm{bp}}/100)^\top Q(x_{\mathrm{bp}}/100) = x_{\mathrm{bp}}^\top Q x_{\mathrm{bp}}/10^4.$$

Four isolated 24/72-optional-name controls retain identical state rows, the linear term, constant and native settings. Both representations return audited portfolios and native integer-optimal labels. Nevertheless unbalanced-graph bounds exceed better audited balanced-graph witnesses by 7.630×10^{-11} and 5.439×10^{-10} original variance, above the unchanged comparison scale. Balanced graphs show no contradiction at that scale. The TE differences are economically small; the diagnostic inconsistency is the relevant mechanism.

Fixed-coordinate elimination can make a group row constant. The compiler removes it only if its coefficients are exactly zero and its constant satisfies both compiled endpoints. Every nonconstant row and constant contradiction remains. Eight graph controls compare keeping/removing such rows, for upper/lower limits just inside and outside the existing audit envelope. In this runtime both graph variants solve the inside-envelope cases and match an independent active-set reference; all outside-envelope variants are infeasible. The historical sizing failure does not reproduce, so no delivery benefit for row elimination is claimed from these controls. Detailed observations are in the supplement.

6. Frozen studies and accounting

All studies use the released OptimalPortfolios 9.2.0 wheel, verified by hash, with QIS 5.44.4, CVXPY 1.9.2, MOSEK 11.2.3 and Clarabel 0.11.1 (Sepp, 2026a,b; Diamond and Boyd, 2016; Goulart and Chen, 2024; MOSEK ApS, 2026). Inputs, numerical source hashes, ordering, seeds and budgets are frozen before each evaluation. No solver is tuned after observing that cohort. AI tools assisted code and manuscript preparation; retained tests, independent references and raw observations support verification.

Original-unit audits retain 10^{-7} NAV feasibility and 10^{-8} trade-zero tolerances. The descriptive TE equality band is 10^{-4} annual bp, separate from feasibility. Comparisons retain failures, open gaps and both signs of TE differences. Native statuses are independently checked on six tiny controls by enumerating 206 supports, with none unresolved; this does not certify larger cases. Primary-fault injections are reported separately from natural fallback.

6.1. Retained 313-case cohort

The ETF case study contains 219 monthly decisions, July 2008–September 2026, from a retrospectively selected 302-ETF universe approved in October 2026. Current books are the previous supplied model weights, normalised and drifted through the intervening month. Each method receives the same current book; methods do not evolve their own holdings recursively.

Targets are an adaptation of the multi-asset allocation framework of Sepp et al. (2026), not a literal replication of its index universe. Their frozen upstream pipeline uses monthly eleven-factor fits, momentum/low-beta signals, corrected log-excess reconstruction, noncash group tracking-variance utility coefficient 10, turnover coefficient 0.4 and 10% instrument caps. Targets were produced with OP 7.21.0, QIS 5.33.1 and FactorLasso 1.0.2 (Sepp, 2026c); execution uses OP 9.2.0. These upstream penalties do not enter (1). The numerical archive supplies targets and risk estimates; public execution replication does not refit the upstream model or reproduce licensed market-data processing.

Execution bands are 0–10% liquidity, 25–45% fixed income, 35–65% equity and 5–15% alternatives, with fractional exposures. Ordinary orders have a 25 NAV bp minimum; equity/alternative holdings have a 50 NAV bp floor. The optional allowance is ten. Only zero to five effective optional candidates remain per decision: this is primarily a preparation/mandatory-execution case study, not evidence of difficult optional search.

The retained synthetic sample has 24 broad cases and two tiny controls. A separately frozen fresh cohort has 64 broad cases and four tiny controls, using two new seeds, diffuse/exit-rich targets, overlapping/disjoint groups and wide/tight funding. Six optional tickets and 144 nominal optional assets already occur in the earlier grid. All retain two compulsory exits, minimum ordinary trade 0.5% and holding floor 0.25%. Historical cases informed prior development; fresh cases did not trigger subsequent tuning. Factorial variants share draws and are not independent market observations.

Independent Exact, Ranked and the actual primary/fallback wrapper are measured. Exact receives ten seconds. Explicit Ranked-started and 60/600-second controls remain separate, with incumbent-generation costs charged. End-to-end timings include preparation but exclude process imports and file reads in this cohort.

6.2. Prospective difficulty and external-risk sample

The new protocol has 18 synthetic cells: 300/1,000/3,000 eligible noncash names, 10/30/100 optional tickets and two predeclared profiles. Current portfolios hold 96 names and 6% cash; targets add 100 names and hold 2% cash. The 0.5% minimum trade and 0.25% holding floor remain feasible. Wide/disjoint/diffuse cases declare two compulsory exits; tight/overlapping/exit-rich cases declare six. Seed, profile and economic features are bundled, with one seed per profile. Nominal size also changes risk draws. These cells are illustrative, not a factorial identification of individual causes or population superiority.

The effective Corridor candidate counts are 155 and 190, with 200–280 ticket states. Nominal universe size, held positions and effective search dimension are therefore reported separately. The same factor-plus-diagonal covariance is supplied in dense form for three representation controls.

Four external-risk cases use OR-Library's 31- and 457-stock panels, at observation cuts 145 and 290 (Beasley et al., 2003; Canakgoz and Beasley, 2009; OR-Library, 2026). Point-in-time QIS log-return EWMA covariance uses span 52 and annualisation 52; artificial weekly dates are assigned because calendar labels are absent. Holdings, targets, cash variance and all execution constraints are synthetic. These are risk-input adaptations, not replications of the original tracking objectives. Owner prices and derived covariance arrays are excluded from redistribution pending reuse terms; source checksums and public regeneration code are supplied. The analytic corridor case completes the 23-case sample.

Main arms are Exact60, Kernel60 and released Ranked with a 180-second external process ceiling. The three tight, 30-ticket cells add Exact10, Exact600, Dense10 and Full60; the analytic case adds Full60. There are 82 declared attempts. All calls run serially, in rotated order, with one solver/numerical library thread and independent starts. Common preparation has a 30-second allowance. Native allowances are not hard wall-clock deadlines: separate process ceilings and all failures are recorded.

This cohort records engine, preparation, representation and whole-process times, including imports and reporting; these differ from the retained cohort's timing scope. A killed worker without a returned portfolio is a failed delivery, even if it held an internal incumbent. Ranked receives no new anytime interface. The prepared base is already audited in every new case, so time to first native feasibility can merely measure acceptance of that start. This is not

a matched anytime-interface experiment. Whole-worker memory includes dense risk reporting; missing killed-worker measurements are not imputed.

7. Results

7.1. Retained decisions and fresh holdout

All 313 cases prepare and all 1,356 arm deliveries pass the independent audit. The actual primary/fallback wrapper uses Exact in all cases and reproduces the independent Exact weights. Nine separate injected failures produce audited Ranked fallbacks. There are no natural fallback observations in this cohort.

Table 2: Retained 313-case cohort: Ranked minus independent Exact TE, in annual bp. Negative differences favour Ranked. The equality band is 10^{-4} bp; the smaller ETF differences remain visible.

Sample	Paired	Ranked lower	Ties	Ranked higher	Minimum	Maximum
ETF	219	12	207	0	-0.000941	0.000043
Existing synthetic	26	0	26	0	0.000000	0.000000
Fresh holdout	68	0	64	4	0.000000	2.457878

Ranked has economically tiny advantages over Exact in 12 ETF cases, up to 0.000941 bp; 207 tie. All 26 existing synthetic cases tie. Exact improves four of 68 fresh cases, with TE differences of 0.288791, 1.926327, 2.457878 and 0.505803 bp in cases 024, 025, 027 and 059. All four have six optional tickets, tight overlapping exposures and audited outputs. They are search misses, not waived constraints. Exact's fresh-cohort median end-to-end time is 0.616 seconds versus Ranked's 4.591; 95th percentiles are 1.935 and 25.609.

The 313 primary calls report 282 native integer-optimal and 31 continuous-optimal statuses. Two near-zero-risk ETF cases exceed the relative-gap target while meeting absolute stopping; the largest relative gap is 0.009671, with absolute gap 5.594×10^{-14} variance. Neither a native label nor the equality band is converted into an independent certificate for the larger cases.

7.2. Larger candidate sets and external risk

All 23 new cases prepare. The 82 attempts yield 64 audited deliveries and 18 process-ceiling failures, all Ranked. Every returned portfolio passes the independent re-audit. Exact60 and Kernel60 deliver in all 18 scaling cells;

Ranked returns in two. Of those two, one ties Exact and one improves it by 2.402490 bp, while both lose to Kernel. Kernel beats Exact60 in 11 cells, ties four and loses three, with differences from −55.718876 to 23.386397 bp.

Table 3: Delivery and process cost in the prospective study. Exact and Kernel have 60-second internal allowances; Ranked retains released defaults with a 180-second process ceiling. Process time includes imports and reporting. Memory is the largest observed worker peak, with its observation count in parentheses; killed workers have no peak measurement.

Sample	Method	Audited	Median s	95th pct. s	Observed MiB
Scaling	Exact	18/18	66.312	67.411	746.137 (18)
Scaling	Ranked	2/18	180.019	180.026	497.750 (2)
Scaling	Kernel	18/18	66.647	67.823	762.730 (18)
External risk	Exact	4/4	35.760	65.471	321.340 (4)
External risk	Ranked	2/4	99.137	180.021	271.801 (2)
External risk	Kernel	4/4	35.840	66.089	348.336 (4)

Table 4: Paired audited TE differences, left method minus Exact60, in annual bp. Negative differences favour the left method. Open Exact gaps mean these are incumbent comparisons. The equality band is 10^{-4} bp.

Sample	Left	Paired	Lower	Ties	Higher	Min bp	Max bp
Scaling	Ranked	2/18	1	1	0	-2.402	0.000
Scaling	Kernel	18/18	11	4	3	-55.719	23.386
External risk	Ranked	2/4	0	2	0	0.000	0.000
External risk	Kernel	4/4	0	2	2	0.000	0.361

The four external-risk cases favour neither the same universal ordering nor an inference of optimality. Exact and Kernel tie on the smaller pair; Exact improves the two larger cases by 0.191694 and 0.361215 bp. Ranked returns only on the smaller pair, where it ties Exact. The optional cap binds in ten of 18 scaling Exact deliveries and 12 of 18 Kernel deliveries. These are incumbent comparisons within the declared domains and resource controls.

Adding the common preparation process gives scaling median research-pipeline times of 72.871 seconds for Exact, 186.578 for Ranked and 73.245 for Kernel. The two import phases mean these are not persistent-production latency estimates. Whole-worker observed peaks reach 746 MiB for Exact

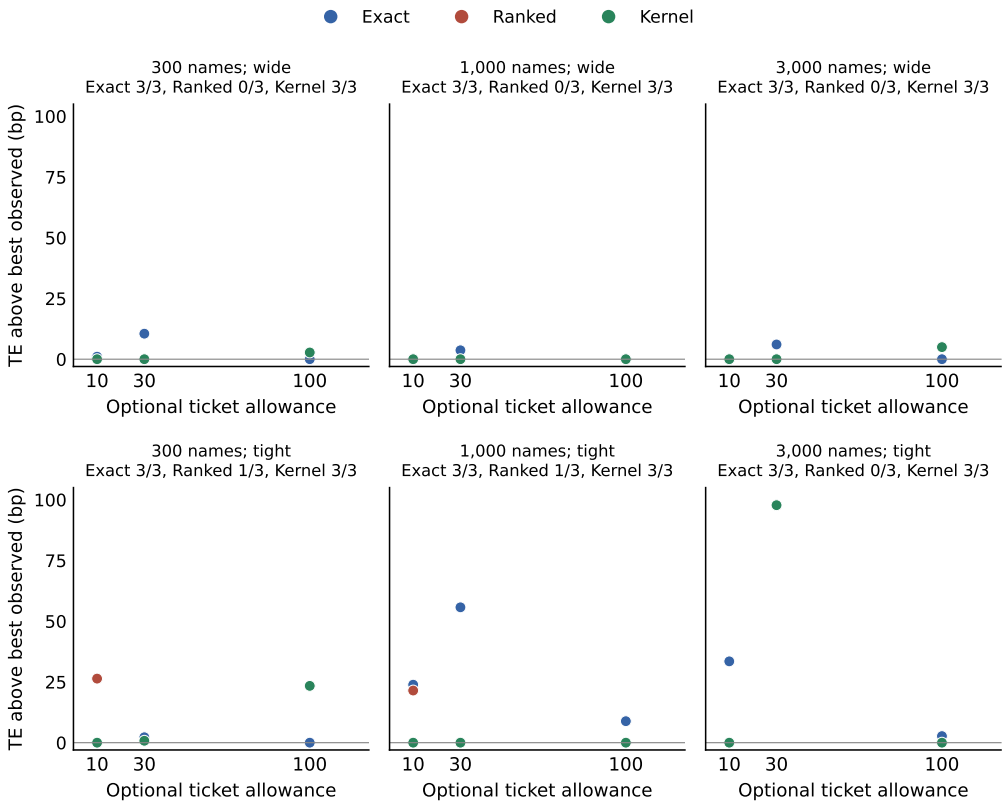


Figure 1: Audited TE above the best observed Corridor delivery, including Exact600 where prespecified. Full-domain results are excluded. Panel counts retain missing deliveries; overlapping points denote similar outcomes. Agreement with an observed incumbent is not proof of optimality.

and 763 MiB for Kernel; this does not establish linear-memory execution or low memory use by killed calls.

7.3. Unresolved gaps and budget controls

Sixteen of 18 scaling Exact60 calls report primal-feasible status; two report integer-optimal status. Median native relative gap is 15.771%, maximum 72.637%. No native bound contradicts an available audited same-domain witness above 10^{-12} variance. That check is weaker than independent certification.

Table 5: All prespecified 30-ticket, tight-profile controls. TE is annual bp. Full60 changes the policy domain; its difference is not search error on the Corridor problem. The final column reports the native relative gap for Exact600, which can differ from the gap of the audited delivery.

Names	Exact10	Exact60	Exact600	Dense10	Full60	Native gap %
300	133.627	92.849	90.613	133.627	133.627	7.554
1000	159.813	159.813	159.813	159.813	159.813	61.837
3000	210.198	210.198	112.441	210.198	210.198	3.170

All three Exact600 controls remain unresolved. Longer computation improves the 300-name case by 2.236112 bp and the 3,000-name case by 97.757191 bp relative to Exact60. The 1,000-name case remains at its prepared base and 61.837% native gap; Kernel60 improves that base by 55.718876 bp. There is no rejected better callback in the Exact600 record: its lack of improvement is not an audit-boundary effect.

Elsewhere the distinction does matter. Twenty native candidate events in scaling Exact60 calls fail audit. In the 300-name, ten-ticket wide case, native incumbent variance is 0.000130181637 but delivered variance is 0.000132001026 after the better vector exceeds the group audit envelope. The native gap therefore understates the accepted portfolio's gap.

Dense10 and structured Exact10 return the same base in all three controls; this does not establish equivalent search paths or memory costs. All three Full60 calls also return the base, yielding one worse incumbent and two ties against Corridor Exact60. Those unfinished comparisons cannot estimate the optimal corridor restriction cost. In contrast, the analytic example returns 100 bp for all Corridor methods and 43.588989 bp for Full, agreeing with the independent scalar solution.

8. Implications and limitations

The execution domain, solver and audit answer separate questions. Corridors restrict economic choices; an unfinished solver limits the search within those choices; numerical acceptance determines whether a returned vector is usable. The counterexample and bound controls show why these distinctions have practical consequences even when two expressions share the same mathematical minimiser.

The evidence supports neither general Ranked superiority nor a universal Exact speed/quality advantage. Kernel's synthetic gains are substantial in several cells, but reverse in others. The tested Exact default is not perspective strengthened, and Ranked's released interface often misses its declared deadline. A new formulation or stopping interface would require separate development and fresh evaluation, not adjustment of this frozen sample.

The ETF universe is retrospective, estimates are supplied and monthly decisions are dependent. Internally generated holdouts are not independent external validation; the external-risk adaptation also has synthetic execution rules. All new bases already pass preparation, so preparation difficulty is not identified. Timings describe one host and declared controls, not universal speed guarantees. Hardware and wall-clock cutoffs can change returned incumbents.

No new primary/fallback arm was run on the difficulty sample. Its standalone failures cannot be converted into an observed fallback frequency, and the wrapper does not guarantee the better of two methods. Nor do these conditional experiments support own-holdings rolling, realised return, Sharpe or transaction- cost claims. The usable result is a reproducible account of policy costs, computational limits and numerical checks, including inconvenient outcomes.

Data and code availability

The companion supplement supplies the full formulation, search order, comparator adaptation and experiment mechanics. The repository replication bundle provides source, protocols, checksums, original-input audits and reference summaries. Separate reviewed archives contain synthetic inputs, raw observations and the supplied de-identified ETF decisions; opaque asset labels provide no guarantee against recognition of numerical histories. Private fund data, identity mappings and vendor histories are excluded. OR-

Library files are retrieved from their owner. Persistent deposit identifiers remain pending. The public code is in OptimalPortfolios (Sepp, 2026a), with QIS analytics (Sepp, 2026b); the exact evaluated wheel and source hashes remain the scientific reference regardless of subsequent API naming changes.

Author declarations

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